

SOUTH AFRICA SOLAR FARM LOCATION PREDICTOR

A Data Science Portfolio Project

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Using data science to identify optimal solar farm locations for underserved rural communities across South Africa

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Executive Summary

This project is set out to identify the minimum number of renewable energy farms needed in South Africa and the optimal locations to build them, using a combination of village-level settlement data, weather observations from the South African Weather Service (SAWS), and almost five years of national grid data from Eskom.

The project was originally designed to cover both solar and wind farm placement. However, following unsuccessful attempts to obtain complete wind data from SAWS, the scope was strategically narrowed to exclusively focus on solar farm placement. This decision was further supported by the availability of pre-computed solar irradiance values in the settlement dataset, making solar suitability scoring more reliable and data driven.

Three datasets were cleaned, merged, and analysed across six structured phases covering exploratory data analysis (EDA), data cleaning, time-series demand forecasting, geospatial suitability scoring, farm count optimisation, and the dashboard. The analysis identified eight priority solar farm locations across three provinces, Northern Cape, Limpopo, and Free State, collectively serving an estimate of 371730 people while covering 59 underserved villages within a 50km transmission radius.

National electricity demand was forecasted using Facebook's Prophet model through to 2030, projecting an average demand of approximately 25346 MW by 2030. A minimum of 221 solar farms would be needed to reach a realistic target of 25% renewable energy penetration by 2030, split across 133 utility-scale installations and 89 small-scale and rooftop solutions targeting constrained areas such as rural KwaZulu-Natal and the Eastern Cape.

Introduction

Background

South Africa faces one of the most pressing energy access challenges on the African continent. Despite being the most industrialised economy in sub-Saharan Africa, millions of citizens, particularly those living in rural and peri-urban settlements, remain without reliable access to electricity. The national grid, operated by Eskom, has historically struggled to meet demand. The situation dramatically worsened over years of widespread loadshedding that peaked between 2022 and 2023, which affected every sector of the economy.

At the same time, South Africa possesses some of the most abundant solar resources in the world. The Northern Cape and Karoo regions receive solar irradiance levels that globally rival the best solar locations, making the country exceptionally well positioned to transition toward renewable energy. The national government has acknowledged this through its Integrated Resource Plan, which sets targets for renewable energy capacity expansion, yet implementation has consistently lagged schedule due to funding constraints, procurement delays, and the complex realities of infrastructure development in a country with significant inequality and geographic diversity.

Project Motivation

This project was motivated by a desire to apply data science to a real-world infrastructure challenge with meaningful consequences for South African communities. Rather than working with a clean, pre-prepared dataset on a well-understood problem, the goal was to tackle the full complexity of a genuine analytical challenge; messy data, incomplete information, domain-specific constraints, and decisions that require both technical rigour and contextual understanding of South Africa's energy landscape.

The project was originally conceived as a solar and wind farm location predictor. Wind energy represents a significant opportunity particularly along the Eastern and Western Cape coastlines and incorporating wind data alongside solar would have produced a more complete renewable energy siting model. Due to unforeseen circumstances, complete wind observation data was never received. Rather than allow this to delay the project indefinitely, a strategic decision was made to narrow the scope to solar farm placement. This decision was supported by the presence of pre-computed solar irradiance values in the village-level settlement dataset, which provided a reliable and granular measure of solar potential across all 76941 settlements analysed.

Project objectives

This project aimed to achieve the following:

- Clean and integrated three heterogeneous dataset covering village-level settlement characteristics, national grid performance, and weather station observations into a unified analytical framework
- Analyse electricity demand patterns across South African provinces to identify where energy access need is greatest
- Forecast national electricity demand through to 2030 using time-series modelling to quantify the future energy gap
- Score all 76941 villages across South Africa for solar farm suitability using a weighted multi-criteria model informed by solar potential, security risk, population need, agricultural land use, and infrastructure accessibility
- Identify the minimum number of solar farms needed to reach a 25% renewable energy penetration target by 2030
- Recommend specific priority locations for solar farm development, ranked by their combined suitability and community impact

Scope and Limitations

This project focuses exclusively on solar farm placement for rural and peri-urban settlements that are either unelectrified or underserved by the existing national grid. It does not address large-scale industrial or urban electricity demand, which is captured separately in Eskom's national grid data and would require a different analytical approach. The recommendations produced by this project are intended as a procurement plan. Land tenure, environmental impact assessments, and detailed grid integration studies would be required before any recommended location could proceed to implementation.

Data Sources

This project integrated three primary datasets, each covering a different dimension of South Africa's energy landscape. The datasets varied significantly in format, granularity, and quality, requiring substantial cleaning and reshaping before analysis could begin.

Eskom National Grid Data

The Eskom dataset provided hourly national grid performance data spanning April 2021 to March 2026, comprising of 43824 rows and 32 columns. The data was obtained directly from Eskom and covered a comprehensive set of generation and demand metrics including residual demand, renewable energy output broken down by type (wind, photovoltaic solar, concentrated solar power, and other sources), installed capacity per generation type, international imports and exports, and both forecasted and actual demand figures.

This dataset served as the foundation for the national demand forecasting component of the project and provided the baseline renewable energy penetration rate of 8.7% against which the optimisation targets were set. One column, the original residual forecast before lockdown, was dropped during cleaning due to it consisting of 65% of missing values. This reflects its limiting applicability to the post-COVID period covered by the analysis.

A notable characteristic of this dataset was a consistent year-on-year decline in average national demand from 24341 MW in 2021 to 21847 MW in 2025. Rather than indicating reduced electricity consumption, this trend reflects the rapid uptake of rooftop solar installations by South African households and businesses following the server loadshedding of 2022 and 2023, which effectively removed a growing portion of electricity demand from Eskom's visibility.

Village-Level Settlement Dataset

The settlement dataset provided the most granular and analytically rich data source in the project, covering 76941 settlement polygons across South Africa with 42 attributes per settlement. The dataset included spatial identifiers such as geohash codes, polygon geometries, and coordinates; demographic information including population estimates and building counts by size category; infrastructure indicators such as distance to the nearest main road, distance to existing electricity transmission lines, and proximity to the nearest urban hub; energy access indicators including nightlight presence, grid

overlap percentage, number of connections, and electricity demand estimates; agricultural land use data including crop types, agricultural area, and yield values; a pre-computed solar irradiance score per village; a relative wealth index; and security risk classifications derived from ACLED conflict event data.

This dataset required the most extensive cleaning of the three sources. Key interventions included correcting a province name typo where Northern Cape was recorded as Nothern Cape across 1955 rows, parsing semicolon-separated fatality counts into structured columns representing four ACLED conflict event types, encoding the security risk categorical variables as an ordinal numeric score, capping extreme outliers at the 99th percentile to prevent urban settlements such as Johannesburg and Cape Town from dominating the model, and imputing 11999 missing relative wealth index values using province-level medians.

South African Weather Service (SAWS) Station Data

Weather data was obtained from SAWS covering eight stations across South Africa for the period January 2021 to February 2026. The data was provided in two Excel files in a multi-block format. This required custom parsing logic to extract and reshape into a usable tidy format before any analysis could begin.

The first file contained daily maximum and minimum temperature readings for all eight stations. The second file contained total cloud cover measured in octas, a scale of 0 to 8 where 0 represents a completely clear sky and 8 represents full cloud coverage, for five of the eight stations. Cloud cover data was recorded three times a day at 08:00, 14:00, and 20:00 and was averaged to a single daily value per station before merging with the settlement data.

The eight stations (Cape Town, Beaufort-Wes, Umthatha, De Aar, Port Nolloth, Mafikeng, Irene, and Thohoyandou) were selected strategically to provide geographic coverage across the provinces identified as most suitable for solar farm development, while deliberately excluding stations in provinces where terrain, agricultural land use, security constraints, or flooding risk made large-scale solar farm development impractical. Three stations, Beaufort-Wes, Port Nolloth, and Thohoyandou, had no cloud coverage data available and were left as null values in the merged dataset rather than being imputed, to avoid introducing potentially misleading cloud cover estimates that could affect solar suitability scoring.

The project was originally designed to incorporate wind speed data from SAWS alongside the temperature and cloud cover measurements. Despite submitting a formal data request and multiple follow-up communications, complete wind observation data was not received within the project timeline. The decision was therefore made to proceed with solar farm placement, supported by pre-computed solar irradiance values available in the settlement dataset.

Methodology

This project followed a structured six-phase analytical pipeline, progressing from raw data ingestion through to actionable farm placement recommendations. Each phase was built directly on the outputs of the previous one, ensuring that every analytical decision was grounded in cleaned, validated data before moving forward.

Phase 1: Data Acquisition and Exploratory Data Analysis (EDA)

The first phase involved loading all three datasets and conducting a thorough exploratory data analysis to understand their structure, quality, and analytical potential before any cleaning or modelling took place.

The Eskom dataset was loaded and inspected for missing values, data types, and temporal coverage. The datetime column was converted from a string format to a proper pandas DatetimeIndex and the data was resampled to confirm continuity across the full five-year period. Initial visualisations revealed four distinct patterns in the national demand data; a long-term declining trend driven by rooftop solar adoption, a strong annual seasonality with demand peaking in June and July during South Africa's winter months, a diurnal pattern with morning and evening consumption peaks corresponding to household activity, and random residual noise consistent with loadshedding events, public holidays, and weather-related demand fluctuations.

The SAWS Excel files were found to be in a complex multi-block format, where each station-year combination was stored as a separate block within a single worksheet, accompanied by legend headers and using non-standard missing value markers including asterisks and dashes. Custom parsing logic was developed to iterate through each row, detect station metadata headers using regular expressions, and extract daily observations into a tidy long format dataframe.

The settlement dataset was profiled to understand the distribution of key variables. This phase revealed that electricity demand was heavily concentrated in KwaZulu-Natal and the Eastern Cape due to the high number of settlement polygons in those provinces rather than higher per-village consumption, that solar irradiance values were highest in the Northern Cape and the surrounding Karoo region, and that 54% of all villages were classified as high risk; a finding with significant implications for farm placement feasibility.

Phase 2: Data Cleaning

Each dataset was cleaned independently before any merging took place, ensuring that cleaning decisions could be made in the context of each dataset's own characteristics without interference from the others.

Eskom cleaning involved dropping the lockdown forecast column due to 65% missing values, removing 20 zero-demand rows that represented recording failures rather than genuine zero consumption, excluding the incomplete 2026 data beyond March to avoid distorting the forecast model, and setting an explicit hourly frequency on the DatetimeIndex to confirm data continuity.

SAWS cleaning was performed using a combination of Python for structured reshaping and SQL for data quality validation and joining. The two Excel files were parsed into a unified long format dataframe, missing value markers were replaced with proper null values, the three daily cloud cover readings per station were averaged to a single daily value, and the data was pivoted from a long to a wide format to produce one row per station per date. SQL was then used to validate temperature ranges, flag the one instance where maximum temperature was recorded below minimum temperature, clip cloud values to the valid 0 to 8 octas range, and join the maximum temperature, minimum temperature, and cloud cover measurements into a single clean table.

Settlement cleaning was performed using a combination of the Data Wrangler extension in Visual Studio Code for visual cleaning tasks and Python for programmatic transformations. Key cleaning steps included correcting the Northern Cape province name typo across 1955 rows, parsing the semicolon-separated ACLED fatality columns into four typed sub-columns per radius plus a total, encoding security risk as an ordinal variable with low equal to 0, medium equal to 1, and high equal to 2, capping outliers at the 99th percentile for demand, population, building counts, and hull area to prevent urban settlements from dominating the model, correcting building density values above 100% which represented measurement artefacts from overlapping polygon boundaries, imputing missing relative wealth index values using province-level medians, and converting Boolean columns to integer format for compatibility with machine learning libraries.

Phase 3: Consumption Trend Analysis and Demand Forecasting

The third phase used the cleaned Eskom dataset to analyse historical demand patterns and forecast future electricity demand through to 2030.

Time-series decomposition was performed using the statsmodels seasonal decompose function on weekly resampled demand data, confirming the presence of all four demand patterns identified during EDA; trend, annual seasonality, diurnal pattern, and white noise residuals. The Augmented Dickey-Fuller test was applied to formally test for stationarity. The initial test returned a p-value of 0.9873, indicating non-stationarity, which was subsequently traced to the presence of 20 zero-demand recording errors and the anomalous 2026 partial year data rather than genuine non-stationarity in the underlying demand process. After removing these observations, the declining year-on-year trend was identified as the primary source of non-stationarity, which was driven by the accelerating adoption of rooftop solar installations following the loadshedding crisis of 2022 and 2023.

Model selection settled on Facebook's Prophet forecasting library rather than a traditional ARIMA or SARIMA model for three reasons. First, Prophet handles multiple seasonality patterns simultaneously without requiring manual specification of differencing parameters. Second, Prophet automatically detects changepoints, which are moments where the trend changes direction or rate. This was particularly relevant given the structural shift demand behaviour caused by the end of widespread loadshedding in 2025. Third, Prophet's built-in support for South African public holidays allowed the model to account for demand dips on days such as Christmas, Easter, and Heritage Day without additional feature engineering.

The model was trained on data from April 2021 to December 2024 and evaluated against a held-out test set covering the full calendar year of 2025. The model achieved a Mean Absolute Percentage Error of 8.56% on the test set, which is within the acceptable threshold of 10%, established prior to modelling given the constraints of a less than five-year training dataset with a partial first year.

Bias correction was applied to address a systematic overestimation of 1773 MW identified on the 2025 test set. This overestimation reflected the model's inability to account for the structural regime change caused by the end of loadshedding and the consequent reduction in the grid dependence by households and businesses that had invested in rooftop solar. A bias correction factor derived from the mean test set error was applied to all future forecasts, producing corrected demand estimates of 22546 MW for 2026, 23242 MW for 2027, 23925 MW for 2028, 24645 MW for 2029, and 25346 MW for 2030.

Phase 4: Geospatial Analysis and Solar Suitability Scoring

The fourth phase merged the cleaned datasets and built a suitability scoring model to rank all 76941 villages for solar farm placement potential.

Dataset merging used a nearest-neighbour spatial join to assign each village to its closest SAWS weather station based on Haversine distance, which is a formula that accounts for the curvature of the Earth when calculating distances between latitude and longitude coordinates. SAWS data was first aggregated from daily readings to station-level summary statistics covering the full 2021 to 2026 period, producing average maximum temperature, average minimum temperature, average temperature range, and average cloud cover per station. These aggregated weather features were then joined to each village based on its nearest station assignment.

Feature selection identified nine variables as inputs to the suitability scoring model, chosen because of their direct relevance to solar farm viability and community impact. Distance to existing transmission lines was explicitly excluded despite its presence in the dataset, on the grounds that the project targets off-grid communities where proximity to the existing grid is not a prerequisite for solar farm variability. The nine selected features were solar irradiance value, average cloud cover, average maximum temperature, electricity demand, population, distance to main road, security risk score, agricultural land value, and the underserved village flag.

Feature normalisation applied Min-Max scaling to transform all nine features to a uniform 0 to 1 range, with directionality applied so that features where lower values indicate better suitability were inverted after scaling so that a score of 1 always represents the most favourable value regardless of the original direction of the variable.

Weight derivation used correlation-based weighting against a composite target variable combine 45% electricity demand, 45% solar irradiance, and 10% underserved village status. This approach was preferred over arbitrary manual weighting because it grounds each feature's influence in its empirically measured relationship with the project's core objective, making the weights defensible to any stakeholder who questions the methodology. The derived weights assigned 32.7% to solar irradiance, 24% to population, 17.9% to cloud cover, 8.1% to security risk, 5% to maximum temperature, 4.6% to agricultural land value, 4% to underserved status, and 3.7% to road distance.

Transmission coverage analysis calculated the number of villages, underserved villages, total population, and total electricity demand falling within a 50km radius of each top-ranked location. The 50km radius was chosen based on the practical transmission distance for small to medium utility-scale solar farms beyond which line losses become significant enough to meaningfully reduce supply reliability.

Final ranking combined the suitability score with transmission coverage metrics using a weighted final score that assigned 40% to underserved villages within 50km, 30% to the suitability score, 20% to the population served, and 10% to the total demand within 50km. This weighting intentionally prioritised community impact over location quality, reflecting the project's core objective of reaching the most underserved communities rather than simply identifying the most technically optimal solar sites.

Phase 5: Optimisation (Minimum Farm Count)

The fifth phase used the demand forecasts from Phase 3 to calculate the minimum number of solar farms needed to close the gap between current renewable energy output and a realistic target penetration level.

The current renewable energy penetration rate of 8.7% was established from the Eskom dataset as the baseline. A target of 25% renewable energy penetration by 2030 was selected as the planning horizon, deliberately conservative relative to the government's stated ambitions to account for South Africa's consistent history of infrastructure procurement delays, funding reallocation, and the practical realities of project approval timelines in the country.

Each utility-scale farm was modelled with an installed capacity of 75 MW and a capacity factor of 25%, reflecting South Africa's solar conditions and producing an average continuous output of 18.75 MW per farm. The minimum farm count was calculated by dividing the energy gap, the difference between the target renewable output and the current renewable output, by the average output per farm, with the result rounded up to the nearest whole number to ensure the target is always met rather than narrowly missed.

The total farm requirement was split into 60% utility-scale farms suitable for flat, low-risk, high-solar-potential provinces such as the Northern Cape, Limpopo, and the Free State, and 40% small-scale and rooftop installations targeting constrained provinces such as KwaZulu-Natal and the Eastern Cape where difficult terrain, high security risk, and agricultural land use make large utility-scale farms impractical. A phased implementation framework is proposed with the eight priority locations identified in Phase 4 forming Phase 1, additional utility-scale farms forming Phase 2, and small-scale and rooftop solutions forming Phase 3.

Phase 6: Dashboard Development

The final phase translated the analytical outputs into an interactive four-page Power BI dashboard designed to communicate findings to both technical and non-technical audiences.

Page 1 provides a national overview including KPI cards for total villages analysed, national demand, underserved village count, underserved percentage, and the average potential, alongside a donut chart showing South Africa's current electricity supply mix relative to the 25% renewable target and a provincial demand bar chart with conditional colour formatting.

Page 2 presents the demand concentration and constraints analysis, including a security risk breakdown by province, a demand versus population scatter chart, and a conditionally formatted table showing demand and security risk by province and district.

Page 3 displays the solar suitability analysis with an interactive map of recommended farm locations, a ranked suitability score bar chart, a solar potential versus suitability scatter chart, and an impact summary table showing transmission coverage metrics for each recommended location.

Page 4 presents the optimisation and forecast outputs, with year-selectable KPI cards, a clustered bar chart comparing utility-scale and small-scale farm requirements by year, and a forecast line chart showing projected demand from 2027 to 2033 with confidence interval bounds.

Key Findings

This section summarizes the most significant insights produced at each phase of the analysis, drawing together the analytical outputs into a coherent narrative about South Africa's solar energy opportunity.

Energy Poverty is Concentrated in Provinces with the Most Challenging Deployment Conditions

EDA revealed that KwaZulu-Natal and the Eastern Cape account for the highest village-level electricity demand in the dataset, with KwaZulu-Natal contributing approximately 118000 MW of aggregated demand across its 28756 settlement polygons. This is nearly double the next highest province. However, these same provinces present the most significant barriers to utility-scale solar farm development, including mountainous and hilly terrain that complicates construction and panel installation, the highest concentration of high security risk settlements where infrastructure theft and vandalism represent ongoing operational costs, and limited flat agricultural land suitable for large solar installations.

This finding has important implications for energy planning. It demonstrates that the provinces with the greatest energy need are not the provinces where large solar farms can most easily be built, and that a single uniform deployment strategy cannot adequately serve all of South Africa's underserved communities. A differentiated approach, utility-scale farms in suitable provinces transmitting electricity to surrounding areas, combined with small-scale and rooftop installations in constrained provinces, is necessary to address energy poverty effectively across the full geographic diversity of the country.

South Africa's Grid Demand is Declining Due to Rooftop Solar Adoption

Analysis of the Eskom national grid data revealed a consistent and accelerating decline in average national demand from 24341 MW in 2021 to 21847 MW in 2025; a reduction of approximately 2494 MW over four years. This trend does not reflect reduced electricity consumption by South African households and businesses. Rather, it reflects the rapid uptake of rooftop solar installations following the severe loadshedding crisis of 2022 and 2023, which prompted a wave of private solar investments that effectively removed a growing share of electricity demand from Eskom's grid.

This finding has two significant implications for the project. First, it introduces a structural regime change that the Prophet forecasting model cannot fully account for, resulting in a systematic overestimation of future grid demand that required explicit bias correction. Second, it suggests that the true unmet electricity demand across South Africa is higher than Eskom's grid data alone indicates, since a portion of demand is now being self-supplied through private rooftop installations that are invisible to national grid measurements. The energy gap that solar farms need to fill is therefore likely larger than the corrected forecast figures suggest, making the project's farm count recommendations conservative rather than overstated.

The Northern Cape Consistently Emerges as South Africa's Premier Solar Province

Three independent lines of evidence converged on the Northern Cape as the highest priority province for utility-scale solar farm development. The solar potential map showed the darkest concentration of high irradiance values centred on Springbok and the surrounding Namaqualand region, with annual pv value scores reaching up to 2074 kWh per square meter. The cloud cover analysis showed De Aar and surrounding Northern Cape stations recording the lowest average cloud cover of all monitored locations, with an average of 2.28 octas indicating predominantly clear skies throughout the year. The security risk analysis showed the Northern Cape has the highest proportion of low security risk villages of any province, making it the most conducive environment for infrastructure development and long-term maintenance.

This convergence of solar resource quality, atmospheric clarity, and security suitability explains why the Northern Cape already hosts South Africa's largest operational solar installations and validates the dataset's consistency with established knowledge about South Africa's renewable energy geography.

Optimal Farm Locations Balance Solar Quality Against Community Impact

The suitability scoring model identified Upington and the surrounding Z F Mgcawu district as the highest scoring locations on pure solar suitability metrics. However, when transmission coverage analysis was incorporated into the final ranking, measuring the number of underserved villages and total population within a 50km transmission radius, Thabazimbi, in Limpopo emerged as the top overall recommendation despite scoring lower on pure suitability.

Thabazimbi's top ranking reflects its position within reach of 108202 people and 10 underserved villages within its 50km transmission radius. This means that it would serve nearly double the

population coverage of the next best location. This finding illustrates a fundamental tension in renewable energy sitting between optimising for resource quality and optimising for community impact. A farm built in the most technically optimal location may serve fewer people than one built in a slightly less optimal location that happens to be surrounded by a larger underserved population. The final ranking methodology deliberately resolves this tension in favour of community impact by assigning 40% of the final score to underserved village coverage, reflecting the project's core objective of reaching communities without electricity access.

Setsoto in the Free State Serves the Most Underserved Villages of Any Recommended Location

AN unexpected finding emerged from the transmission coverage analysis. Setsoto Local Municipality in the Free State, a province that was not originally prioritised for solar farm development due to its characterisation as a predominantly agricultural province, serves 22 underserved villages within its 50km transmission radius. This is the highest underserved village coverage of any location in the top eight recommendations, exceeding even Thabazimbi's coverage of 10 underserved villages.

This finding demonstrates the value of allowing data to challenge assumptions. The decision to exclude Free State from the initial SAWS station selection was based on reasonable domain knowledge about the province's agricultural character, but the settlement data reveals that pockets of significant energy poverty exist within the province that would benefit substantially from a nearby solar installation. Setsoto's inclusion in the final recommendations as the third ranked location reflects this data driven revision of the initial assumption.

Reaching 25% Renewable Energy Penetration by 2030 Requires 221 Solar Farms

The optimisation analysis calculated that closing the gap between South Africa's current renewable energy penetration of 8.7% and a target of 25% by 2030 would require a minimum of 221 solar farms, based on corrected demand forecasts and standard utility-scale farm specifications of 75 MW installed capacity with a 25% capacity factor. This total is split into 133 utility-scale farms suitable for flat, accessible, low-risk provinces and 88 small-scale and rooftop installations targeting constrained provinces.

The 25% target was selected as a realistic, rather than aspirational, planning horizon, deliberately accounting for South Africa's consistent history of infrastructure procurement delays and the extended

approval timelines that characterise large-scale construction projects in the country. A more optimistic target of 35% would require approximately 329 farms in total by 2030. A 35% target is only achievable in theory but inconsistent with the practical realities of infrastructure delivery.

The eight priority locations identified in Phase 4 represent the first phase of a proposed three-phase implementation framework. Phase 1 covers the eight priority farms identified by the suitability model. Phase 2 covers the remaining utility-scale farms in suitable provinces. Phase 3 covers small-scale and rooftop installations in constrained provinces, with a particular focus on rural KwaZulu-Natal and Eastern Cape communities, where utility-scale development is not feasible, but energy poverty is most acute.

Recommendations

Based on the findings of this analysis, the following recommendations are made for solar farm development in South Africa. The recommendations are structured across two levels; specific priority locations identified through the suitability scoring model, and a phased implementation framework for reaching the 25% renewable energy penetration target by 2030.

Priority Solar Farm Locations

The suitability scoring model evaluated all 76941 settlement polygons across South Africa and produced a ranked list of recommended solar farm locations based on a weighted combination of solar irradiance, cloud cover, temperature, population need, security risk, agricultural land value, road accessibility, and underserved village status. The eight deduplicated priority locations and their key metrics are presented below.

Rank	Location	Province	Suitability Score	Underserved Villages within 50km	Population Served	Total Demand within 50km (MW)	Final Score
1	Thabazimbi Local Municipality	Limpopo	0.850	10	108202	39282.8	0.626
2	Kai !Garib Local Municipality	Northern Cape	0.859	6	55336	19132.1	0.533
3	Setsoto Local Municipality	Free State	0.840	22	55494	16824.4	0.532
4	Upington	Northern Cape	0.860	5	42078	14948.4	0.490
5	Gamagara Local Municipality	Northern Cape	0.853	10	42657	12858.4	0.472
6	Tsantsabane Local Municipality	Northern Cape	0.851	5	27512	9866.4	0.302
7	Siyathemba Local Municipality	Northern Cape	0.852	0	9296	2862.6	0.185
8	Emthanjeni Local Municipality	Northern Cape	0.847	1	31155	9179.1	0.178

Several observations are worth highlighting from this ranked list:

- Thabazimbi ranks first despite not being in the Northern Cape
- Setsoto, in the Free State, ranks third despite being an unexpected inclusion
- Gamagara and Tsantsabane form a Northern Cape cluster where the two locations have overlapping 50km transmission radii. A combined deployment in this area would provide continuous coverage across a larger portion of the Northern Cape. The geographic proximity makes them natural candidates for concurrent development
- Siyathemba ranks seventh despite a high suitability score

Phased Implementation Framework

Reaching the 25% renewable energy penetration target by 2030 requires a minimum of 221 solar farms in total. Given the practical realities of infrastructure development in South Africa, a single large-scale deployment is not feasible. Instead, a three-phase implementation framework is proposed, structured to deliver immediate impact through priority locations while progressively scaling toward the 2030 target.

Phase 1: Priority Locations (Immediate Deployment)

Phase 1 covers the eight priority locations identified by the suitability and impact scoring model. These locations were selected because they represent the best available combination of solar resource quality, security conditions, infrastructure accessibility, and community impact across three provinces.

The recommended starting point within Phase 1 is Tsantsabane Local Municipality, in the Northern Cape. Despite ranking sixth overall, Tsantsabane offers the most favourable conditions for an initial deployment; flat terrain, low security risk, good road access, and a location within the overlapping coverage zone with Gamagara that allows a second farm to be added incrementally. Thabazimbi, while ranking first in overall impact, represents a larger and more complex deployment that is better suited to Phase 1's later stages once initial operational experience has been gained.

Location	Province	Priority	Rationale
Tsantsabane	Northern Cape	First	Lowest complexity, flat terrain, overlaps with Gamagara
Gamagara	Northern Cape	Second	Adjacent to Tsantsabane, extends Northern Cape cluster
Upington	Northern Cape	Third	Highest suitability score, established infrastructure
Kai !Garib	Northern Cape	Fourth	Strong solar potential, good road access
Setsoto	Free State	Fifth	Highest underserved village coverage
Thabazimbi	Limpopo	Sixth	Highest population impact, larger development
Emthanjeni	Northern Cape	Seventh	Completes Northern Cape cluster
Siyathemba	Northern Cape	Eighth	Strong solar resource, lower priority due to minimal underserved coverage

Phase 2: Utility Scale Expansion

Phase 2 scales the utility-scale farm programme beyond the eight priority locations, adding the remaining utility-scale farms need to approach the 25% penetration target. The number of additional farms required increases year on year as demand grows.

Year	Total Farms Needed	Phase 1 Farms	Phase 2 Utility Farms	Phase 3 Small Scale Farms
2027	203	8	114	82
2028	208	8	117	84
2029	215	8	121	86
2030	221	8	125	89

Phase 2 farms should be sited using the same suitability scoring methodology applied in this project extended to additional high-scoring locations beyond the top eight. The Northern Cape, Limpopo, and the Free State should remain the primary focus for utility-scale development given their demonstrated advantages in solar resource quality, security conditions, and terrain suitability.

Phase 3: Small Scale and Rooftop Solutions

Phase 3 addresses the energy access challenge in provinces where utility-scale solar farm development is not feasible. KwaZulu-Natal and the Eastern Cape present compounding barriers to large farm deployment; hilly and mountainous terrain that increases construction costs and complexity, high security risk that creates ongoing operational exposure to panel theft and vandalism, significant agricultural land that limits available flat land, and limited road infrastructure in the most remote areas. These constraints do not eliminate the need for solar energy in these provinces, but they do change the appropriate form of delivery.

Small-scale solar installations at the village or household level, and rooftop solar systems for schools, clinics, and community centers, represent a more practical and cost-effective solution for these constrained environments. A single police station or health clinic with rooftop solar can serve as community anchors, to provide electricity for critical services while reducing the security exposure associated with large unattended infrastructure.

Phase 3 targets 80 to 89 small-scale installations across 2027 to 2030 period, focused primarily on the most underserved districts of KwaZulu-Natal and the Eastern Cape including O.R. Tambo, Alfred Nzo, Amathole, and Umkhanyakude; the districts with the highest concentration of underserved villages identified in the settlement dataset.

Additional Deployment Opportunity: Solar Carports for Urban and Peri-Urban Areas

The suitability scoring model and phased implementation framework presented in this report focus primarily on ground-mounted utility-scale solar farms for rural and peri-urban settlements and small-scale rooftop installations for constrained provinces. A fourth deployment category warrants consideration for provinces that fall outside the scope of both approaches; particularly Gauteng.

Gauteng presents a specific challenge for solar energy planning. The province has significant electricity demand, contributing 38000 MW of aggregated village-level demand in this analysis, but limited available flat land for utility-scale solar development, high urbanisation that reduces rooftop availability at the household level, and a density of commercial and industrial activity that makes large ground-mounted installations impractical. Despite these constraints, Gauteng possesses an asset that is abundant, underutilised, and exceptionally well exposed to South Africa's solar resource; commercial and industrial parking infrastructure.

Solar carports, also known as photovoltaic carport structures, involve mounting solar panels on canopy structures over outdoor parking bays at shopping centers, office parks, hospitals, universities, and industrial facilities. This approach generates electricity without requiring additional land beyond what is already developed, provides the added benefit of shade for parked vehicles, and can be implemented through Power Purchase Agreements in which a renewable energy developer funds and installs the system at no cost to the property owner in exchange for a long-term electricity supply contract at rates below Eskom's standard tariff. South Africa already has operational precedents for this model at several major commercial properties including large retail centers in Gauteng and KwaZulu-Natal.

From a transmission perspective, solar carports are well suited to urban environments because their electricity is consumed primarily on site, which reduces the property's draw from the grid without requiring any long-distance transmission. Excess generation can be exported into the local distribution network through net metering or wheeling arrangements, supplying surrounding buildings and residences with minimal transmission losses given the short distances involved within a commercial district.

For provinces such as Gauteng, KwaZulu-Natal, and the Western Cape, where utility-scale solar farm development is constrained by terrain, security risk, or land availability, solar carport programmes targeting major commercial nodes represent a practical and financially viable pathway to increasing renewable energy penetration without the land use conflicts, security exposure, or long-distance transmission challenges associated with utility-scale rural installations. The South African government's existing Section 12B tax incentive, which allows businesses to deduct renewable energy asset costs from taxable income, provides a ready-made financial mechanism to accelerate commercial uptake without requiring direct government capital expenditure.

Key Caveats

These recommendations are intended as a data driven analytical starting point for solar energy planning, not a definitive engineering or procurement plan. Before any recommendation location proceeds to implementation, the following additional assessments would be required:

- Land tenure and ownership – confirming that identified sites are available for development and resolving any competing land use claims

- Environmental impact assessment – evaluating the ecological implications of construction and operation at each site
- Grid integration study – assessing the capacity of existing transmission infrastructure to absorb new generation at each location
- Detailed site surveys – ground-truthing terrain, soil conditions, and access routes at each recommendation location
- Community consultation – engaging with local communities and municipalities to ensure alignment with local development priorities

Limitations

No analytical project is without constraints, and intellectual honesty about the boundaries of a model's validity is as important as the findings themselves. The following limitations are acknowledged as relevant to the interpretation and application of this project's recommendations.

Weather Station Coverage

The SAWS data used in this project only covers eight stations across the country, assigned to villages using a nearest-neighbour spatial join. This means that every village, in a given region receives the same aggregated weather statistics as the nearest station regardless of how far that village is from the station. A village 200km from Cape Town WO receives identical temperature and cloud cover values to one 50km away, which introduces a degree of geographic imprecision into the solar suitability scores for settlements far from any station.

This limitation was unavoidable given the data available. The SAWS station selection was made strategically to maximise coverage of the provinces identified as most suitable for solar development while excluding provinces where development constraints made weather data less relevant to the analysis. Future iterations of this project would benefit from a denser network of weather stations, or from incorporating satellite-derived solar irradiance data such as NASA's POWER dataset which provides gridded daily irradiance estimates at 0.5 degree resolution across South Africa without relying on physical station proximity.

Absence of Wind Data

This project was initially designed to model both solar and wind farm placement. Due to time constraints, the scope was consequently narrowed to solar farm placement.

The absence of wind data means that potentially suitable wind farm locations, particularly along the Eastern and Western Cape coastlines and in the high-altitude interior regions where wind resources are strongest. South Africa's wind resources are complementary to its solar resources in an important way: wind generation tends to be stronger during winter months and at night when solar output is lowest, meaning a combined solar and wind deployment strategy would produce a more consistent and reliable renewable energy supply than solar alone. Incorporating wind data into future iterations of this model would produce a more complete and operationally robust set of recommendations.

Demand Forecast Overestimation

The Prophet forecasting model was trained on Eskom's national grid data from April 2021 to December 2024. This was unfortunately a period characterised by widespread loadshedding and rapidly accelerating rooftop solar adoption. The model achieved a Mean Absolute Percentage Error of 8.56% on the 2025 test set, which is within the acceptable threshold established prior to modelling; which was established as 10%. However, the model systemically overestimated 2025's demand by an average of 1773 MW, reflecting its inability to account for the structural regime change caused by the end of widespread loadshedding and the consequent reduction in grid dependence by households and businesses that had invested in rooftop solar.

A bias correction factor was applied to all future forecasts to address this systemic overestimation. However, the correction assumes that the magnitude of overestimation observed in 2025 will remain constant through to 2030, which may not hold if rooftop solar adoption continues to accelerate; in which case, the corrected forecasts may still overestimate future grid demand. The farm count recommendations derived from these forecasts should therefore be interpreted as an upper bound on the number of utility-scale farms needed rather than a precise minimum, with the actual requirement potentially lower if private solar adoption continues at its current pace.

Population and Demand Capping

Outlier capping at the 99th percentile was applied to population and demand values during data cleaning to prevent large urban settlements from dominating the suitability scoring model. This resulted in all top ranked recommended locations receiving a capped population value of 7422, rather than their actual municipal population figures, which are subsequently higher. Thabazimbi Local Municipality, for example, has an actual population significantly larger than the capped value suggests.

This limitation does not affect the relative ranking of recommended locations since the cap was applied consistently across all 76941 villages, ensuring that no location gained an unfair advantage. However, it does not mean that the population served figures reported in the recommendations table understate the true number of people who would benefit from each farm, and should be interpreted as minimum estimates, rather than precise counts.

Static Suitability Scoring

The suitability scoring model produces a single static score per village based on data aggregated across the 2021 to 2026 period. It does not account for temporal variation in solar resource availability, seasonal fluctuations in demand, or changes in security conditions over time. A village that scores highly today may face different conditions in the next five years.

Future iterations of this model would benefit from incorporating time-varying features that allow the suitability score to be updated as conditions change, producing a dynamic rather than static ranking that reflects the current state of each location at the time a deployment decision is being made.

Exclusion of Industrial and Commercial Demand

The settlement dataset covers rural and peri-urban village-level settlements. It does not capture large-scale industrial, mining, or commercial electricity demand which constitutes a significant portion of the country's total electricity consumption. The demand figures in this analysis therefore only represent the residential and small business consumption of underserved communities and should not be interpreted as a complete picture of provincial or national electricity demand.

The Eskom national grid data provides a complementary view of total national demand but does not disaggregate by province or end-use category in a way that would allow direct integration with the village-level analysis. Combining the two perspectives, village-level residential demand and provincial industrial demand, into a unified demand model would require additional data sources such as NERSA's licensed electricity distribution data and would represent a valuable extension of this work.

Single Transmission Radius Assumption

The 50km transmission radius used to calculate coverage metrics for each recommended farm location was applied uniformly across all eight locations regardless of local grid infrastructure, terrain, or population distribution. In practice, the effective transmission radius of a solar farm depends on the capacity and condition of local distribution infrastructure, the topography between the generation site and demand centers, and the number and size of communities within range.

A more precise coverage analysis would calculate transmission radii on a location-by-location basis using actual grid infrastructure data from Eskom and local municipalities, accounting for terrain-adjusted line distances rather than straight-line Haversine distances. This would produce more accurate estimates of the population and demand that each farm could realistically serve and would strengthen the evidence base for the phased implementation recommendations.

Conclusion

This project set out to answer a question with genuine consequences for millions of South Africans, where should solar farms be built, and how many are needed to meaningfully close the gap between South Africa's current renewable energy output and a realistic near-term target? Six months of analytical work across six structured phases produced a concrete, data-driven answer to both questions.

The analysis identified eight priority solar farm locations across three provinces, the Northern Cape, Free State, and Limpopo, collectively capable of serving an estimated 371730 people and reaching 59 underserved villages within their combined 50km transmission radii. Thabazimbi Local Municipality, Limpopo, emerged as the top overall recommendation not because it has the best solar resource but because it serves the largest underserved population within transmission range, reflecting the project's deliberate prioritisation of community impact over technical optimality. The Northern Cape's dominance of the suitability rankings validated the dataset's consistency with established knowledge about the country's solar geography, while unexpected strength of Setsoto, Free State, demonstrated the value of allowing data to challenge assumptions rooted in domain knowledge alone.

The demand forecasting analysis revealed that South Africa's national grid demand has been declining consistently since 2021; not because South Africans are using less electricity, but because an accelerating wave of private rooftop solar investment following the loadshedding crisis has progressively removed demand from Eskom's visibility. This structural shift introduced a forecasting challenge that required explicit bias correction and serves as a reminder that historical data does not always provide a reliable guide to future conditions when the underlying system has fundamentally changed. The corrected forecasts project average national grid demand reaching approximately 25346 MW by 2030, requiring a minimum of 221 solar farms to achieve 25% renewable energy penetration.

The project also surfaced a finding that extends beyond its immediate analytical scope; that provinces such as Gauteng, KwaZulu-Natal, and the Western Cape, where utility-scale solar farm development faces significant terrain, security, and land use constraints, are not without solar options. Solar carport installations over commercial and industrial parking infrastructure represent a practical, financially viable, and land-efficient pathway to increasing renewable energy penetration in urban environments, complementing the utility-scale and small-scale rural deployment strategies recommended for other provinces.

This project was built on imperfect data. Wind observation data would have enabled a combined solar and wind analysis. Weather station coverage was limited across a geographically diverse country. Population and demand values required capping that understates the true scale of need at recommended locations. These limitations are acknowledged honestly, rather than minimised, because a recommendation that is transparent about its constraints is more useful to a decision maker than one that overstates its precision.

What this project demonstrates above all, is that data science applied thoughtfully to real-world infrastructure challenges can produce insights that are both analytically rigorous and practically meaningful. The eight priority locations identified here are not abstract coordinates on a map, they are specific communities where the combination of solar resource quality, security conditions, infrastructure access, and proximity to underserved populations makes the case for investment compelling and defensible. The phased implementation framework provides a structured pathway from those eight locations to the 221 farms needed by 2030.

South Africa has the solar resource, the technical capability, and the policy frameworks to transform its energy landscape. What has historically been missing is a data-driven, community-centered approach to deciding where to deploy that resource first. This project represents a contribution toward filling that gap.

Technical Appendix

This appendix provides a complete reference for the tools, libraries, datasets, and file outputs produced throughout the project. It is intended for anyone wishing to reproduce, extend, or audit the analytical work documented in this report.

Appendix A: Tools and Technologies

Tool	Version	Purpose
Python	3.13.9	Primary programming language for all data processing, analysis, and modelling
Jupyter Notebook	6.5+	Interactive development environment for phase-by-phase analytical workflow
Visual Studio Code	Latest	Code editor
Data Wrangler	Latest	Visual data cleaning extension used for settlement dataset cleaning
SQLite	3.x	Lightweight relational database used for SAWS data cleaning and joining
Power BI Desktop	Latest	Interactive dashboard development and visualisation
Git	Latest	Version control for all project code and notebooks

Appendix B: Python Libraries

Library	Purpose
pandas	Data loading, cleaning, manipulation, and merging across all datasets
numpy	Numerical operations, log transformations, and array handling
matplotlib	Static visualisation for EDA charts and time-series plots
seaborn	Statistical visualisation for distribution and correlation plots
scikit-learn	Min-Max scaling for suitability score normalisation
statsmodels	Time-series decomposition and Augmented Dickey-Fuller stationarity test
prophet	Demand forecasting model
folium	Interactive map visualisation of recommended farm locations
geopandas	Geospatial data handling and coordinate operations
sqlite3	Python interface for SQLite database operations

re	Regular expression parsing for SAWS multi-block Excel file headers
math	Haversine distance calculation for nearest-neighbour spatial join

Appendix C: Dataset Descriptions

File	Source	Description	Rows	Columns
ESK18239.csv	Eskom	Hourly national grid performance data (April 2021 – March 2026)	43824	32
country stations 2021-26 data1.xlsx	SAWS	Daily maximum and minimum temperature for 8 weather stations (January 2021 – February 2026)	3766	Multi-block
country stations 2021-26 data2.xlsx	SAWS	Daily cloud cover in octas for 5 weather stations (January 2021 – February 2026 at three observation times)	3576	Multi-block
south_africa_dre_atlas_settlements.csv	energydata.info	Village-level settlement characteristics across 76941 South African settlements	76941	42

Appendix D: Cleaned and Derived Output Files

File Name	Phase Produced	Description
eskom_clean.csv	Phase 2	Cleaned Eskom hourly demand data with zero values removed and 2026 partial year excluded
saws_clean.csv	Phase 2	Cleaned and joined SAWS data with one row per station per day, covering max temp, min temp, temp range, and cloud cover
saws_2021_2026.csv	Phase 2	Intermediate long-format

		SAWS data before pivoting, used as SQLite input
saws.db	Phase 2	SQLite database containing saws_raw, saws_wide, and saws_clean tables
settlements_clean.csv	Phase 2	Cleaned settlement dataset with province typo corrected, fatalities parsed, security risk encoded, outliers capped, and missing values imputed
settlements_saws.csv	Phase 4	Merged dataset combining cleaned settlements with nearest-neighbour SAWS weather station assignments and aggregated weather statistics
settlements_saws_scored.csv	Phase 4	Full scored dataset with normalised features, composite suitability score, and derived features for all 76941 villages
settlements_saws_scored_sample.csv	Phase 6	6000 row random sample of scored settlements used for Power BI map visualisation
solar_farm_recommendations_final.csv	Phase 4	Final ranked list of 8 deduplicated priority solar farm locations with suitability scores, transmission coverage metrics, and final impact scores
annual_demand_forecast.csv	Phase 3	Annual summary of bias-corrected demand forecasts for 2026 to 2030
optimisation_summary.csv	Phase 5	Year-by-year optimisation results for 2027 to 2030 including total farms needed, utility-scale and small-scale split, phased implementation counts, and projected renewable energy penetration

Appendix E: SAWS Weather Stations

Station ID	Station Name	Latitude	Longitude	Elevation (m)	Max Temp Data	Min Temp Data	Cloud Data
0021178A3	Cape Town WO	-33.963	18.602	42	Yes	Yes	Yes
0092081 5	Beauford Wes	-32.347	22.573	899	Yes	Yes	No

0127272A4	Umthatha WO	-31.549	28.673	745	Yes	Yes	Yes
0169880 1	De Aar WO	-30.665	23.992	1286	Yes	Yes	Yes
0242644 6	Port Nolloth	-29.250	16.868	8	Yes	Yes	No
0508047 0	Mafikeng WO	-25.803	25.542	1281	Yes	Yes	Yes
0513385A2	Irene WO	-25.910	28.210	1524	Yes	Yes	Yes
0723664 6	Thohoyandou AWS	-23.079	30.383	614	Yes	Yes	No

Appendix F: Suitability Scoring Model (Feature Weights)

Feature	Direction	Derived Weight (%)	Rationale
pv_value	Higher is better	32.71	Primary indicator of solar generation potential
population	Higher is better	24.00	Proxy for the number of people who would benefit from a nearby farm
avg_cloud_octas	Lower is better	17.93	Cloud cover directly reduces solar panel output
security_risk_score	Lower is better	8.11	High security risk increases construction and maintenance costs
avg_max_temp_c	Lower is better	4.98	Thermal degradation reduces panel efficiency above 25°C
ag_value	Lower is better	4.57	High agricultural land value indicates land use competition and tenure risk
is_underserved	Higher is better	4.01	Prioritises villages with limited or no existing electricity access
dist_main_road_km	Lower is better	3.70	Poor road access increases construction and maintenance costs

Weights were derived using correlation-based weighting against a composite target variable combining 45% electricity demand, 45% solar irradiance value, and 10% underserved village status. The distance to existing transmission lines was excluded from the model on the grounds that the project targets off-grid communities for whom proximity to the existing grid is not a prerequisite for solar farm viability

Appendix G: Prophet Model Configuration

Parameter	Value	Rationale
yearly_seasonality	True	Captures winter/summer demand cycle
weekly_seasonality	True	Captures weekday/weekend demand differences
daily_seasonality	False	Disabled because the model operates on daily averages not hourly data
changepoint_prior_scale	0.05	Conservative setting reflecting gradual rather than abrupt trend changes
seasonality_mode	Additive	Appropriate for data without exponential growth
country_holidays	ZA	Accounts for South African public holiday demand dips
training_period	April 2021 to December 2024	Three years of complete annual cycles
test_period	January 2025 to December 2025	Full calendar year covering all seasonal changes
forecast_horizon	2026 to 2030	Five year planning horizon aligned with 25% renewable energy penetration target year
bias_correction	-1773 MW	Applied to correct systematic overestimation identified on the 2025 test set

Appendix H: Optimisation Parameters

Parameter	Value	Rationale
Current RE penetration	8.7%	Derived from Eskom's dataset average across 2021 to 2025
Target RE penetration	25% by 2030	Conservative target accounting for procurement and delivery constraints
Farm installed capacity	75 MW	Standard utility-scale solar PV farm specification for South Africa
Capacity factor	25%	South African average accounting for night, cloud cover, and seasonal variation
Average farm output	18.75 MW	Installed capacity multiplied by capacity factor
Transmission radius	50 km	Practical limit before line losses meaningfully reduced supply reliability
Utility scale proportion	60%	Suitable for flat, accessible, low-risk provinces
Small scale proportion	40%	Targeting constrained provinces where utility-scale development is impractical

Appendix I: Project Notebook Structure

Notebook	Phase	Description
phase1_eskom_data_cleaning.ipynb	Phase 1 and 2	Eskom dataset EDA, cleaning, and exportation of eskom_clean.csv
phase1_saws_data_parsing.ipynb	Phase 1 and 2	SAWS Excel file parsing and reshaping from multi-block format
phase1_saws.ipynb	Phase 1 and 2	SAWS SQL cleaning via SQLite
phase1_settlement_data_cleaning.ipynb	Phase 1 and 2	Settlement dataset EDA, cleaning via Data Wrangler and Python, and exportation of settlements_clean.csv
phase2_settlements_saws.ipynb	Phase 2	Nearest-neighbour spatial join merging settlements and SAWS into settlements_saws.csv
phase3_visualisations.ipynb	Phase 1 to 4	All EDA and analytical visualisations separated to prevent notebook lag
phase3_time_series.ipynb	Phase 3	Prophet model training, evaluation, bias correction, and demand forecast outputs
phase4_geospatial_analysis.ipynb	Phase 4	Suitability scoring, transmission coverage analysis, and farm recommendations
phase5_optimisation.ipynb	Phase 5	Minimum farm count optimisation and phased implementation framework
phase6_sample.ipynb	Phase 6	Random sampling for Power BI map visual, as well as file type conversions from CSV to Excel

Cleaning notebooks were maintained as separate files per dataset to ensure each cleaning pipeline could be inspected, rerun, or modified independently without affecting other datasets. Visualisations were separated into a dedicated notebook to prevent rendering overhead from slowing down analytical notebooks during modelling.

Appendix J: Reproducibility Notes

To produce this analysis, the following steps are required:

1. Install all Python libraries listed in Append B using pip or conda. Ensure SQLite3 is available. Install Power BI Desktop from the Microsoft website to access the dashboard.

2. Place all raw data files listed in Appendix C in a single working directory before running any notebooks. All notebooks assume the raw data files are in the same directory as the notebooks themselves.
3. Run the three cleaning notebooks independently and in any order since they operate on separate datasets with no dependencies between them:
 - Run `phase1_eskom_data_cleaning.ipynb` to produce `eskom_clean.ipynb`
 - Run `phase1_saws_data_parsing.ipynb` and `phase1_saws.ipynb` (requires SQLite3) to produce `saws_clean.csv` and `saws.db`
 - Run `phase1_settlement_data_cleaning` to produce `settlements_clean.csv`
4. Once all three cleaning notebooks have been run successfully and their output files confirmed in the working directory:
 - Run `settlements_saws.ipynb` to produce `settlements_saws.csv` and `settlements_saws_scored.csv`
5. Run `phase3_visualisations.ipynb` at any point after the merging step is complete. This notebook loads `settlements_saws.csv`, `saws_clean.csv`, and `eskom_clean.csv` and produces all EDA and analytical visualisation files saved as PNG and HTML outputs.
6. Run `phase3_time_series.ipynb` after the Eskom the Eskom cleaning notebook has been run. This notebook depends on `eskom_clean.csv` and produces `annual_demand_forecast.csv`. The Prophet library must be installed before running this notebook (installation instructions are available at facebook.github.io/prophet).
7. Run `phase4_geospatial_analysis.ipynb` after `settlements_saws.ipynb` has been run. This notebook depends on `settlements_saws_scored.csv` and produces `solar_farm_recommendations_final.csv`. This notebook contains Haversine distance calculations for transformation coverage analysis which may take several minutes to complete, depending on hardware specifications.
8. Run `phase5_optimisation.ipynb` after `phase3_time_series.ipynb` has been run. This notebook depends on `annual_demand_forecast.csv` and produces `optimisation_summary.csv`.
9. Run `phase6_sample.ipynb` to produce `settlements_saws_scored_sample.xlsx` and other Excel format conversions required for Power BI. This notebook depends on `settlements_saws_scored.csv`.
10. Open `SA_Solar_Dashboard.pbix` in Power BI Desktop. If prompted to refresh data sources, navigate to the working directory containing all Excel output files. Do not load CSV files directly into Power BI, this ensures correct data type detection. The dashboard is pre-configured with all measures, relationships, bookmarks, and formatting applied.

11. All random sampling operations used a fixed random seed of 42 to ensure that the same 6000 village sample is selected every time `phase6_sample.ipynb` is run. All other analytical steps are deterministic and will produce identical outputs on every run, given the same input files.

Notes on Data and Assumptions

The 50km transmission radius applied in the suitability and coverage analysis represents a conservative modelling assumption for distribution-level solar farm transmission. This figure was not derived from a single published source but reflects practical considerations for small to medium utility-scale solar installations where line losses at distribution voltage levels become significant beyond short to medium distances. This assumption is acknowledged as an area for refinement in future iterations of the model.

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